Status	Finished
Started	Saturday, 3 October 2026, 1:40 PM
Completed	Saturday, 3 October 2026, 1:45 PM
Duration	4 mins 57 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Dynamic programming

b. Greedy method

c. Backtracking

d. Divide and conquer

The correct answer is:
Divide and conquer

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element

b. Each part is sorted independently

c. The parts are ignored

d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Searching
b. All of the above
c. Duplicates
d. Selection

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Bubbling up the largest element to its correct position
c. Dividing the list into halves
d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It slows down the process
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Quick Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()
b. arrange()
c. order()
d. sorted()

The correct answer is: sort()

Question **9**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort cannot handle duplicate elements
c. Bubble Sort is less efficient for large lists
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **10**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Arranging data in a particular format
c. Inserting data into a list
d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **11**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Dividing the input into parts, solving each part, and combining the solutions
b. Sorting data sequentially
c. Rearranging data without sorting
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Sorting the entire list sequentially
c. Dividing the input into smaller subproblems
d. Comparing each element with the others

The correct answer is: Dividing the input into smaller subproblems

Question **14**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sorted()
b. order()
c. sort()
d. arrange()

The correct answer is: sorted()

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It makes the search operation slower
- c. It has no effect on the search operation
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching